ModelIt!

Teacher's Guide

Can You Outsmart a Disaster?

How Science Helps Communities Survive Natural Hazards

Quick Reference

Grade Level	Grade 6
NGSS Standard	MS-ESS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS3-2

Analyze and interpret data on natural hazards to forecast future catastrophic events and inform the development of technologies to mitigate their effects.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how hazard intensity, population density, warning time, and building design interact to determine disaster outcomes.
Disciplinary Core Idea	**ESS3.B Natural Hazards** — Mapping the history of natural hazards in a region, combined with an understanding of related geologic forces, can help forecast the locations and likelihoods of future events.
Crosscutting Concept	**Cause and Effect** — Students identify causal relationships between preparation levels and disaster outcomes — more warning time CAUSES fewer casualties, better building design CAUSES less structural damage.

CAST Testing Alignment

MULTIPLE CHOICE:

Two cities of equal population both experience a magnitude 7.0 earthquake. City A has modern building codes while City B has older construction. City A suffers 50 casualties while City B suffers 5,000. Which factor best explains this 100x difference?

CONSTRUCTED RESPONSE:

A city is located on a river floodplain and also near a fault line. The city council has $5 million for disaster preparation. Using evidence from natural hazard data, explain how you would recommend they spend the money and why.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Natural hazards have shaped human civilization throughout history. Modern science and engineering allow us to predict, prepare for, and mitigate the impacts of natural disasters — saving millions of lives each year.

Types of Natural Hazards

Major categories include: geologic hazards (earthquakes, volcanic eruptions, tsunamis, landslides), meteorological hazards (hurricanes, tornadoes, floods, droughts), and wildfires. Each type has specific patterns, warning signs, and mitigation strategies. Understanding these patterns is the first step in disaster preparedness.

The Role of Data

Scientists use historical data, geological surveys, weather monitoring, and satellite imagery to forecast natural hazards. Earthquake recurrence intervals help predict when and where quakes will strike. Hurricane models track storms days in advance. Flood maps identify vulnerable areas. This data drives preparation decisions that save lives.

Engineering for Resilience

Modern engineering provides powerful tools against natural hazards. Earthquake-resistant buildings use base isolation (floating foundations) and flexible materials. Levees and flood walls protect against flooding. Storm shelters withstand tornado-force winds. Fire-resistant materials and defensible space reduce wildfire risk. Each solution addresses a specific hazard.

Japan vs. Haiti: A Case Study

The 2010 Haiti earthquake (M7.0) killed approximately 230,000 people. The 2011 Japan earthquake (M9.0 — 1000x stronger) killed approximately 20,000, mostly from the tsunami. The difference: Japan has strict building codes, tsunami warning systems, regular earthquake drills, and massive infrastructure investment. Haiti had none of these. Preparation is the difference between tragedy and resilience.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — **Locate the System**
Students identify hazard intensity, population density, warning time, and building design as the key components determining disaster outcomes.

E — **Establish Relationships**
Students determine that warning time negatively affects casualties (more warning = fewer deaths), building design negatively affects property damage (better design = less damage), and population density positively amplifies all impacts.

V — **Visualize & Model**
Students build a model showing how preparation factors interact with hazard severity to determine overall disaster impact.

E — **Evaluate Outcomes**
Students run prepared vs. unprepared vs. extreme scenarios to quantify how much difference preparation makes.

R — **Revise & Extend**
Students add emergency response time or community education to explore additional factors in disaster resilience.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Natural disasters happen everywhere. But they don't affect everywhere equally. Why?"*

Watch For

Show side-by-side images of earthquake damage in Haiti vs. Japan. Ask: Why the difference?

Slide 2: Learning Objectives

3 min

Say: *"Today we're figuring out how science and engineering can outsmart disasters."*

Watch For

Have students read objectives. Pre-teach 'resilience,' 'mitigation,' and 'early warning system.'

Slide 3: Big Question

3 min

Say: *"Same size earthquake. One city loses 200 people. Another loses 200,000. What explains the difference?"*

Watch For

Quick write: Students list factors they think make a community more disaster-resilient.

Slide 4: LEVER Framework

2 min

Say: *"We'll model what factors matter most — and which investments save the most lives."*

Watch For

Preview LEVER steps. Connect to real disaster data they'll analyze.

Slide 5: Activity 1: Components

8 min

Say: *"What factors determine whether a community survives a natural disaster?"*

Watch For

Guide sorting. Discuss why hazard intensity and warning time are external (nature and technology), while population density and building design are community choices.

Slide 6: Activity 2: Connections

8 min

Say: *"More warning time saves lives. Better buildings save BOTH lives and property. But more people in harm's way multiplies everything."*

Watch For

Discuss the multiplier effect of population density. Use concrete numbers to illustrate.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test: Does a prepared community survive a disaster that destroys an unprepared one?"*

Watch For

Students predict then run all three scenarios. Compare outcomes quantitatively.

Slide 8: Discoveries

5 min

Say: *"You can't stop a hurricane. But you CAN decide how much damage it does."*

Watch For

Show Haiti vs. Japan data. Connect model results to real-world outcomes.

Slide 9: STEM Challenge

12 min

Say: *"You're now urban planners. You have $10 million and two hazards to prepare for. Go!"*

Watch For

Distribute budget sheets and hazard data. Groups must justify every spending decision with model evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Hazard Intensity, Warning Time
- **INTERNAL:** Population Density, Building Design

Relationships

- **Warning Time to Casualties:** NEGATIVE (-) — More warning time allows evacuation, reducing the number of people in harm's way. However, warning time doesn't protect buildings or infrastructure.
- **Building Design to Property Damage:** NEGATIVE (-) — Better-engineered buildings withstand more force before failing. Modern earthquake-resistant buildings can survive events that would destroy older construction.
- **Population Density to Total Impact:** POSITIVE (+) — Higher population density means more people and structures exposed to the hazard. The same disaster in a dense city affects far more people than in a rural area.

Key Discoveries

- Warning time dramatically reduces casualties but NOT property damage — you can evacuate people but not buildings
- Building design is the most effective factor for reducing BOTH casualties and property damage
- Population density multiplies the impact — the same earthquake in a rural area vs. a dense city produces vastly different outcomes
- No amount of preparation can eliminate all risk from extreme natural hazards — but it can reduce impact by 80-90%

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Natural disasters are unpredictable and random"	While we can't predict the exact time of an earthquake, we CAN identify high-risk zones using historical data and geological mapping. Hurricane tracks are forecast days in advance. Flood zones are well-mapped. Most natural hazard risk is predictable and plannable.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Technology can prevent natural disasters"	Technology can't PREVENT earthquakes, hurricanes, or floods — these are natural Earth processes. But technology can predict them (seismographs, weather satellites), warn about them (alert systems), and reduce their impact (resilient engineering). Mitigation, not prevention, is the goal.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"All disasters affect everyone equally"	Disasters disproportionately affect vulnerable populations — those in poverty, elderly, disabled, and communities with poor infrastructure. A hurricane causes vastly different impacts in a wealthy neighborhood vs. a low-income area. Social and economic factors multiply natural hazard impacts.